SUPPLEMENTAL MATERIAL

Crystal structure of β-porphyranases

The crystal structures of PorA and PorB explain the unambiguous substrate specificity and allow identifying key residues for porphyran recognition in β-porphyranases (For data statistics see Supplementary Table 1). Although quite divergent (27% of sequence identity), both β-porphyranases adopt the family GH16 typical jelly-roll fold and possess the catalytic residues E-139 (mutated against S-139) and E-144 in PorA and E-156 and E-161 in PorB. The structure of the inactive mutant PorA_E139S in complex with L6S-G-L6S-G shows the reducing end bound to the subsites[1] -1 and -2 and that two sugar units extend beyond the substrate binding cleft (Fig. 2a).

Enzyme/substrate complexes in which the substrate spans the cleavage point are difficult to obtain in glycoside hydrolases[2,3]. This is due to the deformation of the pyranose ring at the site of cleavage during catalysis. As a consequence, enzymes evolved to interact with the transition state and bind less the ground state substrate conformation. Furthermore relatively low affinity of the subsite + 1, often observed in glycoside hydrolases, contributes to the lack of binding of oligosaccharides across the point of cleavage.

At the cleavage subsite -1, the G unit forms a stacking interaction with W-131 (W-139, PorB) and hydrogen bonds to residues R-59 (R-70, PorB), D-141 (D-158, PorB) and E-234 (E-256, PorB). The presence of G units is conserved in the primary structure of agarose and porphyran and these sugars are indeed almost super imposable at subsite -1 in PorA and the agarase AgaA[4]. Thus structural differences for substrate recognition predominantly concern the subsite -2, which binds a L6S unit in PorA and a LA unit in AgaA[4]. The superimposition of a L6S unit and a LA unit in the binding cleft of β-agarase A is illustrated in Supplemental Figure 7a.

In PorB the sulfate recognition of L6S at subsite -2 differs with respect to PorA, since H-53 and R-133 are not conserved. The H-53 of PorA is substituted by G-64 in sequence and spatially by the side chain of Tyr-111. However this tyrosine residue does not interact with the sulfate group of L6S. The role of H-53 is carried out in PorB by R-252 that is at the correct distance to form a salt bridge with the sulfate of L6S. This interpretation is supported by a bound morpholino-ethane-sulfonic-acid molecule (MES-buffer) in the subsite -2 of PorB, mimicking the L6S residue. The sulfonic group of the MES molecule is oriented towards the interior of the cleft and interacts with R-252 (Supplementary Fig. 7b). This shows that also in PorB the basically charged pocket is present to harbor the L6S sulfate group as described for PorA. In conclusion, the crystal structures of PorA and PorB explain the unambiguous specificity of porphyranases for the sulfated porphyran from *Porphyra spp*.. Furthermore the crystal structures of two divergent porphyranases allowed to pinpoint the possible variation and to define the key residues for porphyran recognition. This was exploited for the following phylogenetic analysis of environmental porphyranase-like genes.

Supplemental references

1 G. J. Davies, K. S. Wilson, and B. Henrissat, Nomenclature for sugar-binding subsites in glycosyl hydrolases, *Biochem. J.* **321**, 557 (1997).

2 G. J. Davies, S. P. Tolley, B. Henrissat et al., Structures of oligosaccharide-bound forms of the endoglucanase V from Humicola insolens at 1.9 angstrom resolution, *Biochemistry* **34**, 16210 (1995).

3 G. J. Davies, L. Mackenzie, A. Varrot et al., Snapshots along an enzymatic reaction coordinate: Analysis of a retaining beta-glycoside hydrolase, *Biochemistry* **37**, 11707 (1998).

4 J. Allouch, W. Helbert, B. Henrissat et al., Parallel substrate binding sites in a beta-agarase suggest a novel mode of action on double-helical agarose, *Structure* **12**, 623 (2004).

Supplementary Table 1. **Data reduction, phasing and refinement statistics**

Data statistics MAD PorA (ID23-1)	**Peak**	**Edge**	**Remote**
Wavelength (Å)	1.03965	1.04005	1.03530
f' f''	-15.5 13.4	-19.8 9.17	-12.7 10.1
Resolution (Å)	49.9-1.45	49.9-1.45	49.9-1.45
Total Data	167078	160093	148827
Unique Data	82329	78319	72117
Redundancy[a]	2.03 (1.76)	2.05 (1.77)	2.06 (1.85)
Completeness (%)[a]	87.3 (73.4)	85.5 (66.4)	71.2 (36.7)
I/sigI[a]	12.7 (3.08)	12.5 (2.69)	10.9 (2.05)
Rsym (%)[ab]	3.8 (23.8)	4.1 (31.4)	5.1 (42.7)
Phasing (ShelXE)			
CCano (%)	68.2	53.5	58.4
CCiso (%)	76.2	63.7	
FOM[c] before/after DM (to 1.5 Å)	0.22/0.62		

Native data statistics	**PorA complex**	**PorB**
Resolution (Å)	49.3-1.8	44.8-1.8
Total Data	180872	235397
Unique Data	27062	62907
Redundancy[a]	6.7 (6.2)	3.7 (3.7)
Completeness (%)[a]	94.6 (91.4)	100.0 (100.0)
I/sigI[a]	20.8 (8.4)	11.0 (3.7)

Rsym (%)[ab]	5.7 (19.5)	8.9 (33.9)
Rpim (%)[ad]	2.5 (10.3)	5.3 (20.3)

Refinement	**PorA complex**	**PorB**
Resolution range (Å)	49.3-1.8	44.8-1.8
Rwork (Rfree)[e]	15.9 (20.5)	16.1 (20.1)
RMSD bonds [f](Å)	0.022	0.028
RMSD angles (°)	1.97	2.0

Overall B-factors	**PorA complex**	**PorB**
All atoms	13.5	16.2
Protein	11.5	15.1
Ligand/Ions	19.2	16.5
Water	21.7	21.9

N of atoms	**PorA complex**	**PorB**
Protein	2135	4343
Ligand/Ions	76	68
Water	324	446

[a] Values in parenthesis correspond to the highest resolution shell.

[b] $R_{sym} = \Sigma |I-I_{av}| / \Sigma I$, where the summation is over all symmetry equivalent reflections.

[c] FOM, figure of merit.

[d] R_{pim} corresponds to the multiplicity weighted R_{sym}

[e] R-factor calculated on 5% of data excluded from refinement.

[f] RMSD, Root Mean Square Deviation.

Supplementary Table 2: **Genes extracted from Genbank used for Phylogenetic tree construction.**

label	Definition (GenBank)	Taxonomy	GenBank ID
Aga_Aeromo	Beta-agarase	*Aeromonas sp.*	AAF03246
Bp1670	hypothetical protein BACPLE_01670	*B. plebeius* DSM 17135	ZP_03208036
Bp1689	Hypothetical protein BACPLE_01689	*B. plebeius* DSM 17135	ZP_03208055
Aga_Cellvi	putative agarase	*Cellvibrio sp.* OA-2007	BAH16616
GFO_1714	Secreted glycosyl hydrolase, family 16	*Gramella forsetii* KT0803	CAL66684
GFO_1715	Membrane or secreted glycosyl hydrolase, family 16	*Gramella forsetii* KT0803	CAL66685
GFO_1725	Glycosyl hydrolase, family 16	*Gramella forsetii* KT0803	CAL66695
Aga_Maga	Agarase	*Microbulbifer agarilyticus*	BAE06228
Aga_Melo	Agarase	*Microbulbifer elongatus*	BAC99022
Aga_Mthe	Agarase	*Microbulbifer thermotolerans*	BAD29947
MS116	Putative beta-agarase precursor	*Microscilla sp.* PRE1	NP_116804
MS132	Putative beta-agarase	*Microscilla sp.* PRE1	NP_116820
Aga_Patl	Beta-agarase I	*Pseudoalteromonas atlantica* ATCC 19262	AAA91888
Patl0824	Glycosyl hydrolase, family 16	*Pseudoalteromonas atlantica* T6c	YP_660406
CgkA_Pc	Kappa-carrageenase	*Pseudoalteromonas carrageenovora* ATCC 43555	CAA50624
AgaPKJ2-4	Agarase	*Pseudoalteromonas sp.* KJ 2-4	AAR87712
Aga1PND137	Agarase	*Pseudomonas sp.* ND137	BAD88713
Aga2PND137	Agarase	*Pseudomonas sp.* ND137	BAB79291
CgkA_Rb	Kappa-carrageenase	*Rhodopirellula baltica* SH 1	NP_865103
AgaBSd2-40	Agarase	*Saccharophagus degradans* 2-40	YP_526649
AguB	AguB	Uncultured bacterium	AAP49346
AguF	AguF	Uncultured bacterium	AAP49324
Zg4203AgaA	Beta-agarase A precursor	*Zobellia galactanivorans* DSIJ	AAF21820
Zg3573AgaB	Beta-agarase B precursor	*Zobellia galactanivorans* DSIJ	AAF21821
Zg236CgkA	Kappa-carrageenase precursor	*Zobellia galactanivorans* DSIJ	AC27890

Supplementary Table 3. ***Bacteriodes plebeius*** **genes upstream and downstream of the porphyranase are listed against their best Blast Hits found by Genbank + the genome of** ***Zobellia galactanivorans*****.** Further homologue proteins in *Bacteroides thetaiotamicron* and *Bacteroides vulgatus* are presented when existent in their genomes. Blue boxes: the genes with highest identity to genes in *Microscilla* or *Zobellia*. Yellow boxes: Synteny between genes in *B. plebeius* and other gut Bacteroides.

B. plebeius DSM 17135	Strand	Annotation	Best Blast Hit versus (GenBank + *Z. galactanivorans*)	*B. thetaiotaomicron* VPI-548	*B. vulgatus* ATCC 8482
BACPLE_01702	-	L-fucosidase, family GH29	BACCOPRO_00400 (61,3%, *B. coprophilus*)	BT_3665 (34%)	BVU_0217 (35%)
BACPLE_01701	-	sulfatase	Zg332 (56,1%, *Z. galactanivorans*)	BT_3051 (40%)	BVU_0538 (33,8%)
BACPLE_01700	-	Alcohol dehydrogenase	BACCOP_02718 (75,9%, *B. coprocola*)	BT_0535 (28%)	BVU_0760 (73,6%)
BACPLE_01699	-	two-component system sensor histidine kinase/response regulator	BACCAC_03608 (33,9%, *B. caccae*)	BT_2923 (28,9%)	BVU_0512 (28,5%)
B. plebeius DSM 17135	Strand	Annotation	Best Blast Hit versus (GenBank + *Z. galactanivorans*)	*B. thetaiotaomicron* VPI-548	*B. vulgatus* ATCC 8482
BACPLE_01698	-	TonB-depedent receptor	Zg3570 (48,5%), Zg3637 (42%, *Z. galactanivorans*)	BT_3332 (32,5%)	BVU_1369 (32,3%)
BACPLE_01697	-	SusD-like protein	Zg3571 (37,9%), Zg3638 (33%, *Z. galactanivorans*)	BT_3025 (27,8%)	BVU_1712 (30,1%)
BACPLE_01696	-	conserved lipoprotein containing CBM22	Zg3572 (36,4%), Zg3639 (28%, *Z. galactanivorans*)	No homologue	No homologue
BACPLE_01695	-	Hypothetical protein	No homologue	No homologue	No homologue
BACPLE_01694	-	β-agarase, family GH86	MS130 (47,9%, *Microscilla sp*. PRE1)	No homologue	No homologue
BACPLE_01693	-	β-agarase, family GH86	MS130 (51,5%, *Microscilla sp*. PRE1)	No homologue	No homologue
BACPLE_01692	-	Conserved hypothetical protein	BF2161 (30,6%, *B. fragilis* NCTC 9343)	No homologue	No homologue
BACPLE_01691	-	Hypothetical protein	No homologue	No homologue	No homologue
BACPLE_01690	-	Hypothetical protein	No homologue	No homologue	No homologue

B. plebeius DSM 17135	Strand	Annotation	Best Blast Hit versus (GenBank + *Z. galactanivorans*)	*B. thetaiotaomicron* VPI-548	*B. vulgatus* ATCC 8482
BACPLE_01689	-	β-porphyranase, family GH16	Zg3640 (49,1%, *Z. galactanivorans*)	No homologue	No homologue
BACPLE_01688	-	Hypothetical protein	No homologue	No homologue	No homologue
BACPLE_01687	+	Hypothetical protein	No homologue	No homologue	No homologue
BACPLE_01686	-	Hypothetical protein	No homologue	No homologue	No homologue
BACPLE_01685	-	Conserved hypothetical protein	BACCOP_00861 (28,3%, *B. coprocola*)	No homologue	No homologue
BACPLE_01684	-	Conserved hypothetical protein	BVU_0514 (73,4%, *B. vulgatus*)	BT_3687 (69,9%)	BVU_0514 (73,4%)
BACPLE_01683	-	Conserved hypothetical protein	BACINT_01451 (45,4%, *B. intestinalis*)	No homologue	No homologue
BACPLE_01682	-	Conserved hypothetical protein	EEI93196 (48,3%, *Sphingobacterium spiritivorum*)	BT_2422 (30%)	No homologue

B. plebeius DSM 17135	Strand	Annotation	Best Blast Hit versus (GenBank + *Z. galactanivorans*)	*B. thetaiotaomicron* VPI-548	*B. vulgatus* ATCC 8482
BACPLE_01681	-	Hypothetical protein	No homologue	No homologue	No homologue
BACPLE_01680	-	Conserved hypothetical protein	BT_3666 (73,4%, *B. thetaiotaomicron*)	BT_3666 (73,4%) / BT_1276 (73%)	BVU_0214 (70,1%) / BVU_0190 (69%)
BACPLE_01679	-	Oxidoreductase	BACDOR_00330 (76,5%, *B. dorei*)	BT_3614 (75,2%)	BVU_0219 (75,8%)
BACPLE_01678	-	Conserved hypothetical protein	BACFIN_03900 (76,9%, *B. finegoldii*)	BT_3615 (77,9%)	BVU_0220 (75,2%)
BACPLE_01677	-	Fucose permease	BACFIN_03899 (76,8%, *B. finegoldii*)	BT_3616 (77,5%)	BVU_0221 (75,9%)
BACPLE_01676	-	Sorbitol dehydrogenase	BT_3617 (75,7%, *B. thetaiotaomicron*)	BT_3617 (75,7%)	BVU_0222 (73,1%)
BACPLE_01675	-	Altronate oxidoreductase	BT_0825 (81,2%, *B. thetaiotaomicron*)	BT_0825 (81,2%)	BVU_3075 (80,6%)

B. plebeius DSM 17135	Strand	Annotation	Best Blast Hit versus (GenBank + *Z. galactanivorans*)	*B. thetaiotaomicron* VPI-548	*B. vulgatus* ATCC 8482
BACPLE_01674	-	Altronate hydrolase	PARMER_00240 (70,7%, *ParaB. merdae*)	BT_0486 (68,2%)	BVU_3053 (68,6%)
BACPLE_01673	-	β-galactosidase, family GH2	Zg4655 (39,1%, *Z. galactanivorans*)	BT_4667 (29,9%)	BVU_2162 (30,1%)
BACPLE_01672	-	β-galactosidase, family GH2	Zg4655 (50,1%, *Z. galactanivorans*)	BT_0461 (29,9%)	BVU_2162 (33,5%)
BACPLE_01671	-	MS110-like protein	Zg4663 (68,8%, *Z. galactanivorans*), MS110 (67,% *Microscilla sp*. PRE1)	No homologue	No homologue
BACPLE_01670	-	β-agarase, family GH16	Zg3573 (agaB, 52,8%, *Z. galactanivorans*)	No homologue	No homologue
BACPLE_01669	-	two-component system sensor histidine kinase Nterminal domain	BVU_0682 (51,5%, *B. vulgatus*)	BT_2619 (27,4%)	BVU_0682 (51,5%)
BACPLE_01668	-	two-component system sensor histidine kinase Cterminal domain	BVU_0682 (59,6%, *B. vulgatus*)	BT_2619 (38%)	BVU_0682 (59,6%)

B. plebeius DSM 17135	Strand	Annotation	Best Blast Hit versus (GenBank + *Z. galactanivorans*)	*B. thetaiotaomicron* VPI-548	*B. vulgatus* ATCC 8482
BACPLE_01667	-	two-component system response regulator	BVU_0681 (58,6%, *B. vulgatus*)	BT_2618 (44,6%)	BVU_0681 (58,6%)
BACPLE_01666	+	Conserved hypothetical protein	PARMER_01229 (52,5%, *ParaB. merdae*)	No homologue	BVU_0680 (51,1%)
BACPLE_01665	+	Mobilization protein C-terminal domain	BACSTE_02978 (92%, *B. stercoris*)	BT_2614 (91,4%)	BVU_0679 (91,2%)
BACPLE_01664	+	Reverse transcriptase	BACUNI_01785 (99,6%, *B. uniformis*)	BT_2297 (61,6%)	BVU_2134 (62,6%)
BACPLE_01663	+	Mobilization protein Nterminal domain	BT_2614 (84,4%, *B. thetaiotaomicron*)	BT_2614 (84,4%)	BVU_0679 (81,1%)
BACPLE_01662	+	Mobilization protein	BACOVA_00482 (74,1%, *B. ovatus*)	BT_2613 (73,6%)	BVU_0678 (72,6%)

Supplementary Table 4. **Primers used for cloning of the five novel GH16 enzymes from *Zobellia galactanivorans* and for the mutagenesis of PorA**

Abbrevation	Cloning strategy	Primer Forward	Primer Reverse
Zg2600 GH16 (PorA)	BamH1/EcoRI	ggggggggATCCCAATTACCATCTCCTACAAACGGG	CCCCCCgAATTCTTAGTCAACCAATTTATACACCCGTACC
Zg3640 (PorE)	BamH1/EcoRI	ggggggggATCCCAGACGCCGCCGCCGCCGG	CCCCCCgAATTCTTATTGATCTACAGGAAGCAAGGTATAC
Zg1017 (PorB)	BamH1/EcoRI	ggggggggATCCCAAGAAGCTCCACATTTTAAGCCTG	CCCCCCgAATTCTTAATTCTTTGAATCAACCAATTGCCATG
Zg3376 (PorC)	BamH1/EcoRI	ggggggggATCCTGTAGCAATTCGGGGGATAATGGT	CCCCCCgAATTCTTACAAATCTTCAAGTTGCCATACCCTA
Zg3628 (PorD)	BglII/EcoRI	gggggggAgATCTCAAGAGCCCCCTAAAACCTATAGT	CCCCCCgAATTCTTATTTGGTCAAATGGATATAGACCGTAC
PorA_E139S	mutagenesis	ccgcgttggccaattttcttcaatcgatgttatcgagcatattgg	ccaatatgctcgataacatcgattgaagaaaattggccaacgcgg

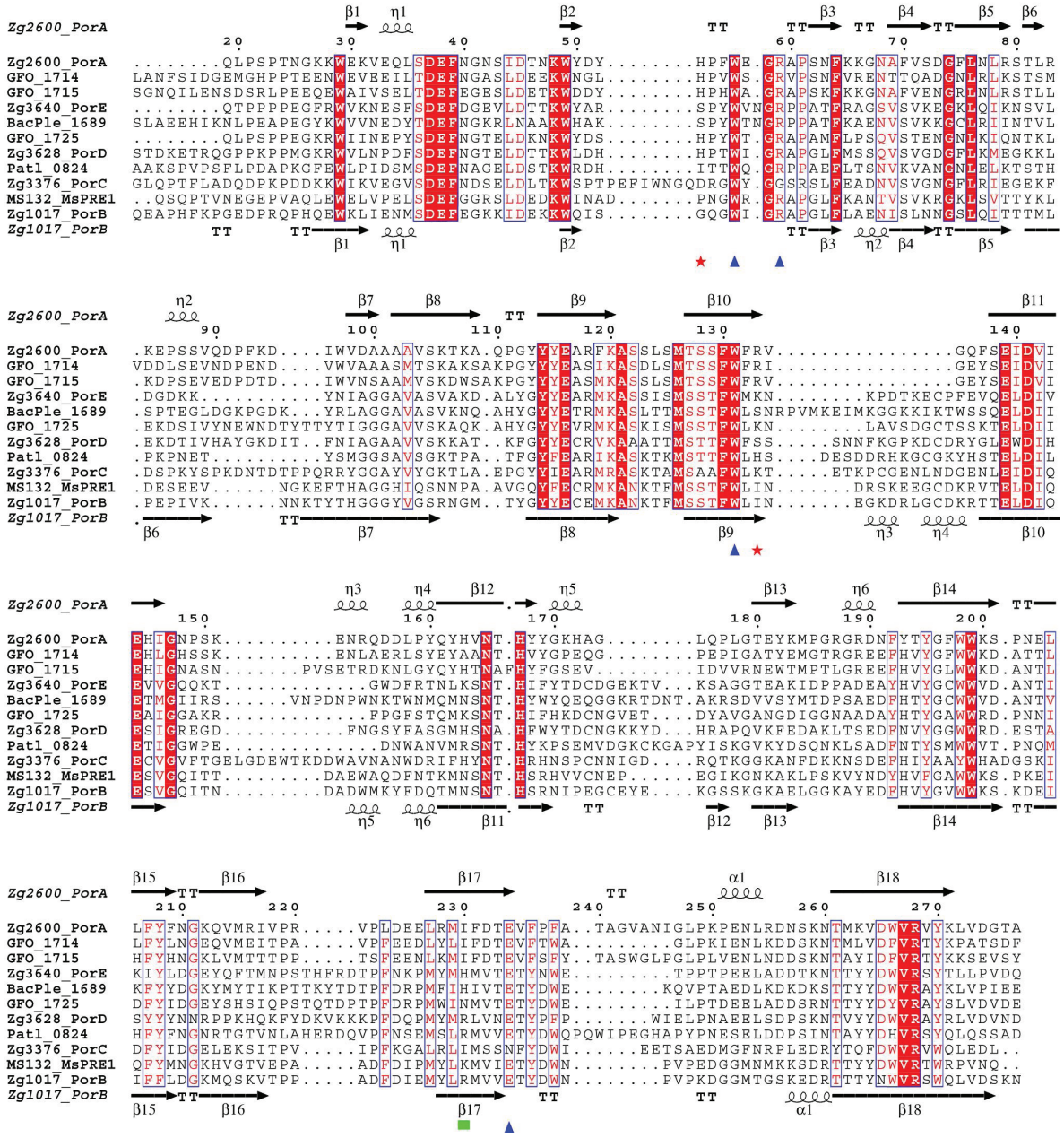
Supplementary Table 5. **Details concerning the origin of Bacteroides species for which genome data are available**

Species name	Accession number	Nationality of volunteer (where known)	Reference	*Source of bacterial strain*	Authors of study	Research facility where study was carried out
Bacteroides caccae	ATCC 43185	American	Int. J. Syst. Bacteriol. **36** (1986), 499-501	*Bacteroides caccae* sp. nov., *Bacteroides merdae* sp. nov., and *Bacteroides stercoris* sp. nov. Isolated from Human Feces	John L. Johnson, W. E. C. Moore and Lillian V. H. Moore	Department of Anaerobic Microbiology, College of Agriculture and Life Sciences, Virginia Polytechnic Institute and State University, Blacksburg, Virginia 24061
Bacteroides capillosus	ATCC 29799	American	Int. J. Syst. Bacteriol. **29** (1979), 427-434	Proposal of Neotype Strains for Seven Non-Saccharolytic *Bacteroides* Species	Elizabeth P. Cato, Lillian V. Holdeman and W. E. C. Moore	Anaerobe Laboratory, College of Agriculture and Life Sciences, Virginia Polytedenic Institute and State University, Blacksburg, Virginia 24061
Bacteroides cellulosilyticus	DSM 14838	French	Int. J. Syst. Evol. Microbiol. **57** (2007), 1516-1520	*Bacteroides cellulosilyticus* sp. nov., a cellulolytic bacterium from the human gut microbial community	Céline Robert, Christophe Chassard, Paul A. Lawson and Annick Bernalier-Donadille	Unité de Microbiologie, INRA, Centre de Recherches de Clermont Ferrand – Theix, 63 122 Saint Genès – Champanelle, France and Department of Botany and Microbiology, University of Oklahoma, Norman, OK 73019, USA
Bacteroides coprocola	DSM 17136	Japanese	Int. J. Syst. Evol. Microbiol. **55** (2005), 2143-2147	*Bacteroides plebeius* sp. nov. and *Bacteroides coprocola* sp. nov., isolated from human faeces	Maki Kitahara, Mitsuo Sakamoto, Masako Ike, Shinji Sakata and Yoshimi Benno	Microbe Division/Japan Collection of Microorganisms, RIKEN BioResource Center, Wako, Saitama 351-0198, Japan
Bacteroides coprophilus	DSM 18228	Japanese	Int. J. Syst. Evol. Microbiol. **57** (2007), 1323-1326	*Bacteroides coprophilus* sp. nov., isolated from human faeces	Hidenori Hayashi, Kensaku Shibata, Mohammad Abdul Bakir, Mitsuo Sakamoto, Shinichi Tomita and Yoshimi Benno	Microbe Division/Japan Collection of Microorganisms, RIKEN BioResource Center, 2-1 Hirosawa, Wako, Saitama 351-0198, Japan and Department of Life Science, Faculty of Agriculture, Tamagawa University, 6-1-1 Tamagawa-Gakuen, Machida, Tokyo 194-8610, Japan
Bacteroides dorei		Japanese	Int. J. Syst. Evol. Microbiol. **56** (2006), 1639 - 1643	*Bacteroides dorei* sp. nov., isolated from human faeces	Mohammad Abdul Bakir, Mitsuo Sakamoto, Maki Kitahara, Mitsuharu Matsumoto, and Yoshimi Benno	Microbe Division/Japan Collection of Microorganisms, RIKEN BioResource Center, Wako, Saitama 351-0198, Japan

					and Laboratory of Dairy Science and Technology, Kyodo Milk Industry Co. Ltd, Hinode, Tokyo 190-0182, Japan
Bacteroides eggerthii	DSM 20697	American (hawaiian individual from Japanese ascendancy)	Int. J. Syst. Bacteriol. **24** (1974), 260-277	New Genus, Coprococcus, Twelve New Species, and Emended Descriptions of Four Previously Described Species of Bacteria from Human Feces	Lillian V. Holdeman and W. E. C. Moore	Anaerobe Laboratory, Virginia Polytechnic Institute and State University, Blacksburg, Virginia 24061
Bacteroides finegoldii	DSM 17565	Japanese	Int. J. Syst. Evol. Microbiol. **56** (2006), 931 - 935	*Bacteroides finegoldii* sp. nov., isolated from human faeces	Mohammad Abdul Bakir, Maki Kitahara, Mitsuo Sakamoto, Mitsuharu Matsumoto, and Yoshimi Benno	Microbe Division/Japan Collection of Microorganisms, RIKEN BioResource Center, Wako, Saitama 351-0198, Japan and Laboratory of Dairy Science and Technology, Kyodo Milk Industry Co. Ltd, Hinode, Tokyo 190-0182, Japan
Bacteroides fragilis		French	Arch. Med. Exp. Anat. Pathol. **10** (1898), 517- 545.	Recherches sur quelques microbes strictement anaérobies et leur role en pathologie	Veillon, A., and A. Zuber	France
			Int. J. Syst. Bacteriol., **26** (1976), 230 - 237	Reinstatement of Species Rank for Bacteroides fragilis, B. ovatus, B. distasonis, B. thetaiotaomicron, and B. vulgatus: Designation of Neotype Strains for Bacteroides fragilis (Veillon and Zuber) Castellani and Chalmers and Bacteroides thetaiotaomicron (Distaso) Castellani and Chalmers	Elizabeth P. Cato and John L. Johnson	Anaerobe Laboratory, Virginia Polytechnic Institute and State University, Blacksburg, Virginia 24061
Bacteroides intestinalis	DSM 17393	Japanese	Int. J. Syst. Evol. Microbiol. **56** (2006), 151 - 154	*Bacteroides intestinalis* sp. nov., isolated from human faeces	Mohammad Abdul Bakir, Maki Kitahara, Mitsuo Sakamoto, Mitsuharu Matsumoto, and Yoshimi Benno	Microbe Division/Japan Collection of Microorganisms, RIKEN BioResource Center, Wako, Saitama 351-0198, Japan and Laboratory of Dairy Science and Technology, Kyodo Milk Industry Co. Ltd, Hinode, Tokyo 190-0182, Japan
Bacteroides ovatus	ATCC 8483	American	J. Bacteriol. **25** (1933), 389-413	The bacteroides of human feces.	Eggerth, A. H., and B. H, Gagnon	Department of Bacteriology, Hoagland Laboratory, Brooklyn
			Int. J. Syst. Bacteriol., **26** (1976), 230 - 237	Reinstatement of Species Rank for *Bacteroides fragilis*, *B. ovatus*, *B. distasonis*, *B. thetaiotaomicron*, and *B. vulgatus*: Designation of Neotype Strains for *Bacteroides fragilis* (Veillon and Zuber) Castellani and Chalmers and *Bacteroides thetaiotaomicron* (Distaso) Castellani and Chalmers	Elizabeth P. Cato and John L. Johnson	Anaerobe Laboratory, Virginia Polytechnic Institute and State University, Blacksburg, Virginia 24061
Bacteroides pectinophilus	ATCC 43243	American	Appl. Environ. Microbiol. **52** (1986), 880-887	*Bacteroides pectinophilus* sp. nov. and *Bacteroides galacturonicus* sp. nov.: Two Pectinolytic Bacteria from the Human Intestinal Tract	N S Jensen and E Canale-Parola	Department of Microbiology, University of Massachusetts, Amherst, Massachusetts 01003

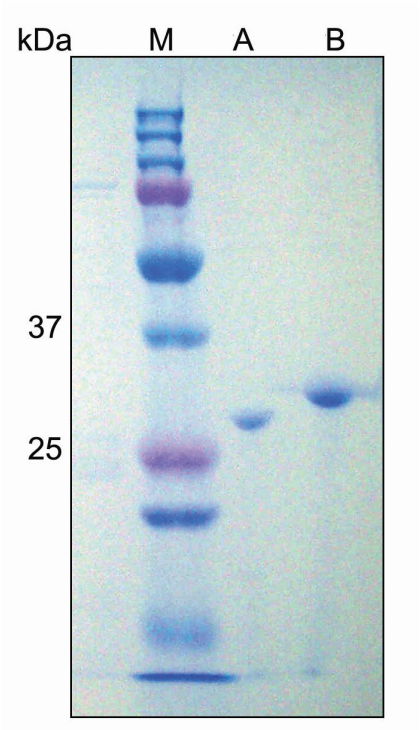
Bacteroides plebeius	DSM 17135	Japanese	Int. J. Syst. Evol. Microbiol. **55** (2005), 2143-2147	*Bacteroides plebeius* sp. nov. and *Bacteroides coprocola* sp. nov., from human faeces	Maki Kitahara, Mitsuo Sakamoto, Masako Ike, Shinji Sakata and Yoshimi Benno	Microbe Division/Japan Collection of Microorganisms, RIKEN BioResource Center, Wako, Saitama 351-0198, Japan
***Bacteroides* sp.**	1_1_6	n/a	Unpublished		Washington University (USA)	
***Bacteroides* sp.**	2_1_7	n/a	Unpublished		Washington University (USA)	
***Bacteroides* sp.**	2_2_4	n/a	Unpublished		Washington University (USA)	
***Bacteroides* sp.**	3_2_5	n/a	Unpublished		Washington University (USA)	
***Bacteroides* sp.**	4_3_47FAA	n/a	Unpublished		Washington University (USA)	
***Bacteroides* sp.**	9_1_42FAA	n/a	Unpublished		Washington University (USA)	
***Bacteroides* sp.**	D1	n/a	Unpublished		Washington University (USA)	
***Bacteroides* sp.**	D2	n/a	Unpublished		Washington University (USA)	
Bacteroides stercoris	ATCC 43183	American	Int. J. Syst. Bacteriol. **36** (1986), 499-501	*Bacteroides caccae* sp. nov., *Bacteroides merdae* sp. nov., and *Bacteroides stercoris* sp. nov. Isolated from Human Feces	John L. Johnson, W. E. C. Moore and Lillian V. H. Moore	Department of Anaerobic Microbiology, College of Agriculture and Life Sciences, Virginia Polytechnic Institute and State University, Blacksburg, Virginia 24061
Bacteroides thetaiotaomicron	VPI-5482	American	J. Bacteriol. **25** (1933), 389-413	The bacteroides of human feces.	Eggerth, A. H., and B. H, Gagnon	Department of Bacteriology, Hoagland Laboratory, Brooklyn
			Int. J. Syst. Bacteriol., **26** (1976), 230 - 237	Reinstatement of Species Rank for *Bacteroides fragilis*, *B. ovatus*, *B. distasonis*, *B. thetaiotaomicron*, and *B. vulgatus*: Designation of Neotype Strains for *Bacteroides fragilis* (Veillon and Zuber) Castellani and Chalmers and *Bacteroides thetaiotaomicron* (Distaso)	Elizabeth P. Cato and John L. Johnson	Anaerobe Laboratory, Virginia Polytechnic Institute and State University, Blacksburg, Virginia 24061
Bacteroides uniformis	ATCC 8492	American	J. Bacteriol. **25** (1933), 389-413	The bacteroides of human feces.	Eggerth, A. H., and B. H, Gagnon	Department of Bacteriology, Hoagland Laboratory, Brooklyn
Bacteroides vulgatus	ATCC 8482	American	J. Bacteriol. **25** (1933), 389-413	The bacteroides of human feces.	Eggerth, A. H., and B. H, Gagnon	Department of Bacteriology, Hoagland Laboratory, Brooklyn
			Int. J. Syst. Bacteriol. **26** (1976), 230 - 237	Reinstatement of Species Rank for *Bacteroides fragilis*, *B. ovatus*, *B. distasonis*, *B. thetaiotaomicron*, and *B. vulgatus*: Designation of Neotype Strains for *Bacteroides fragilis* (Veillon and Zuber) Castellani and Chalmers and *Bacteroides thetaiotaomicron* (Distaso) Castellani and Chalmers	Elizabeth P. Cato and John L. Johnson	Anaerobe Laboratory, Virginia Polytechnic Institute and State University, Blacksburg, Virginia 24061

Supplementary Figures

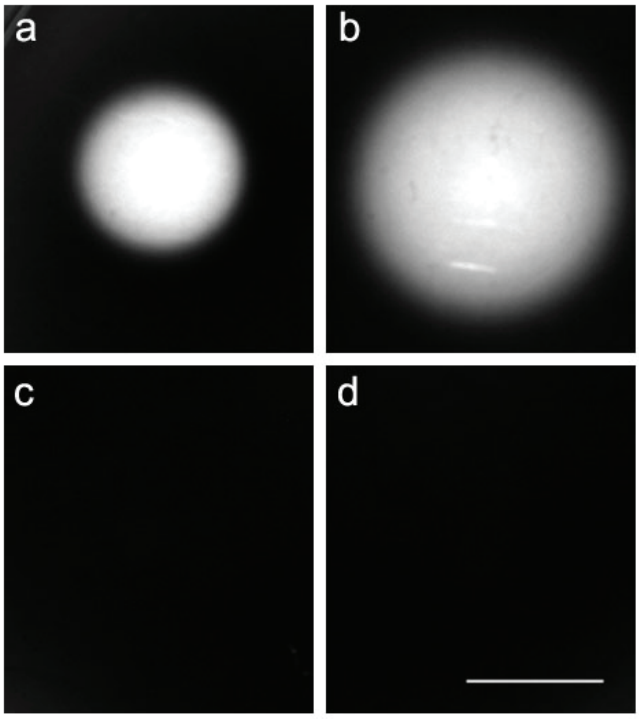


Supplementary Figure 1. **Structural alignment of GH16 proteins here identified as β-porphyranases that are found in marine bacteria and in the gut bacterium *B. plebeius*** (BacPle_1689) that was isolated from a Japanese individual. The sequences are compared with the secondary structures of Zg2600 (PorA above) and Zg1017 (PorB below). Alpha helices and beta strands are represented as helices and arrows, respectively, and beta turns are marked with TT. The blue triangles marked the residues involved in the recognition of the D-galactose in the subsite -1

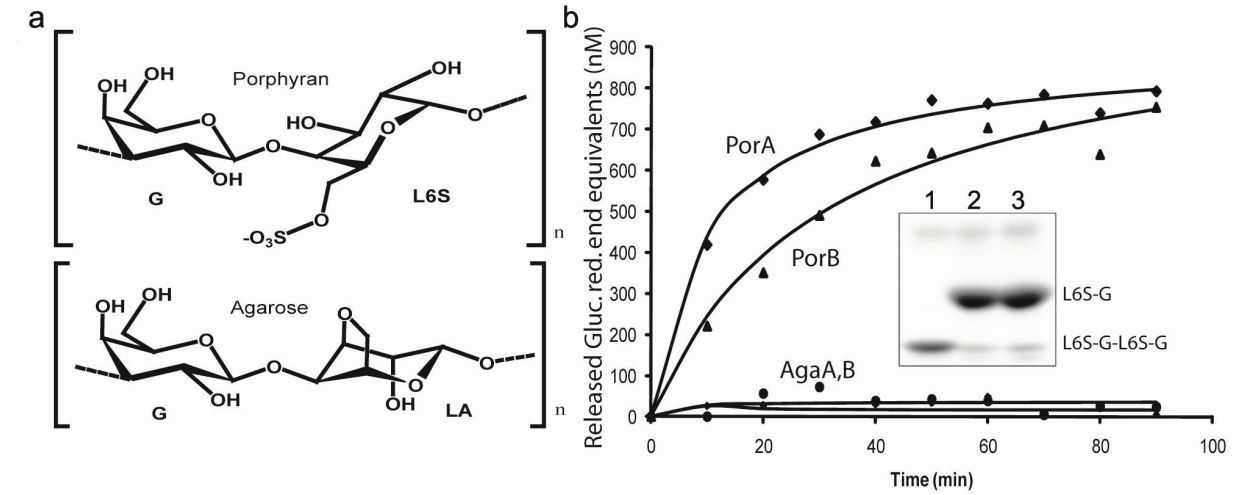
of PorA and PorB. The residues involved in the recognition of the L-galactose 6-sulfate in subsite -2 of PorA and PorB are marked by red stars and green squares, respectively. This sequence alignment was prepared using ESPript[39] with the sequences listed in the Supplementary Table 2.



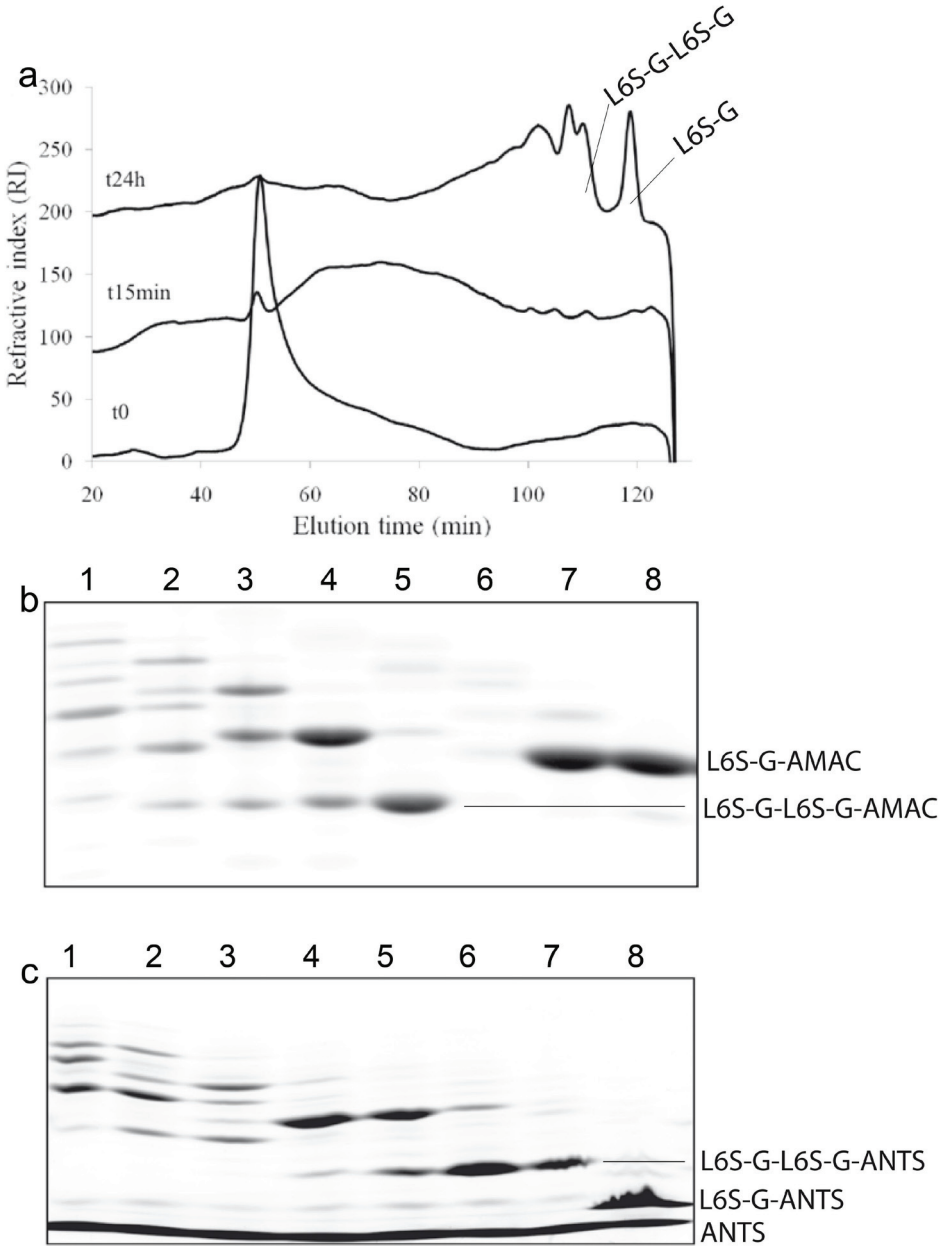
Supplementary Figure 2: **SDS-PAGE analysis of recombinant family GH16 enzymes from *Zobellia galactanivorans* produced in *Escherichia coli* BL21(DE3)**. **M**: Protein marker. **A**: The purified family GH16 catalytic module of the protein Zg2600 (PorA). **B**: The purified protein Zg1017 consisting of a family GH16 module only (PorB).



Supplementary Figure 3. **β-Porphyranases are not active on agarose gel compared to β-agarases. (a)** 1 µg of AgaA, **(b)** 1 µg of AgaB, **(c)** 1 µg of PorA, **(d)** 1 µg of PorB were placed on an agarose plate and incubated overnight at 30 ºC. Activity was revealed with Lugol and is shown by a bright halo. Inset scale length = 1cm.

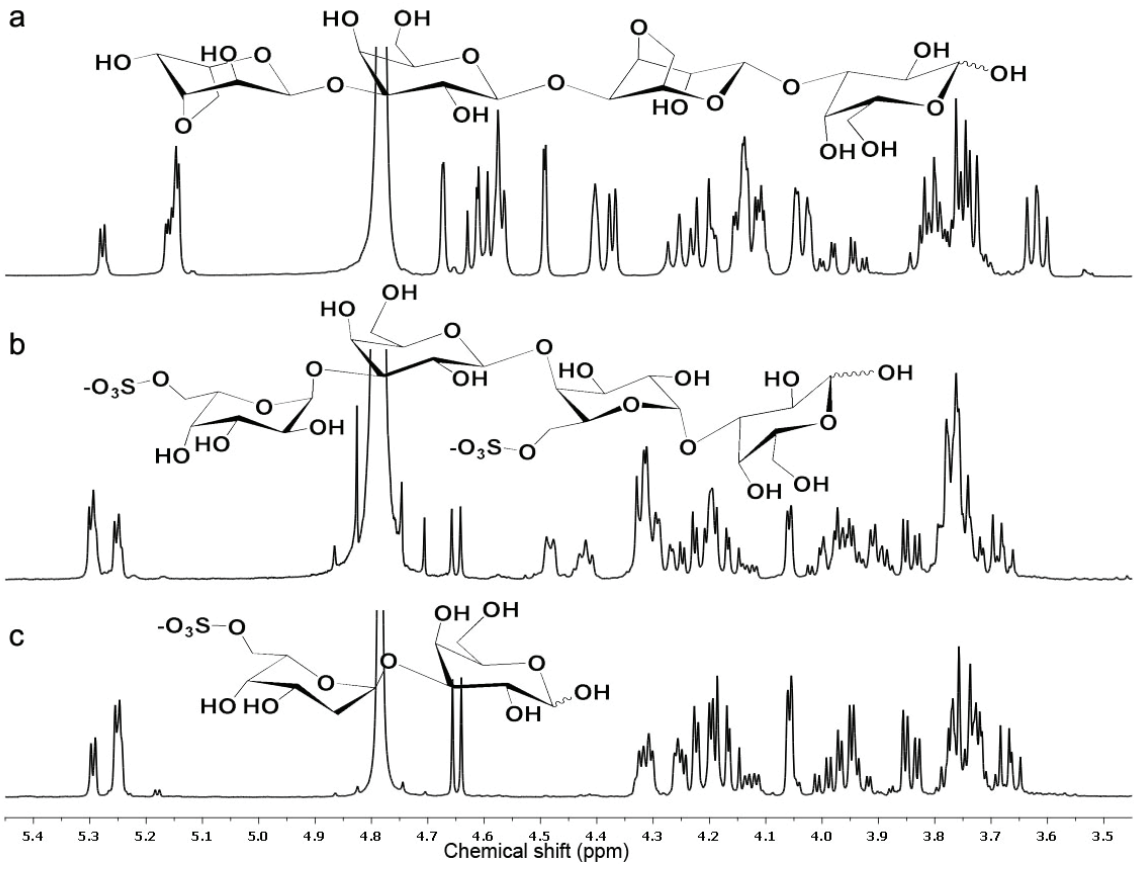


Supplementary Figure 4: **PorA and PorB are β-porphyranases. (a)** Structure of the porphyran polysaccharide repeat unit in comparison with agarose. **(b)** Activity of the β-porphyranases PorA and PorB (~5 nM) on pure porphyran (0,125% w/v) compared with the β-agarases AgaA and AgaB. **Inset:** The porphyran tetrasaccharide is degraded to disaccharide by both porphyranases as shown with fluorophore labelled reaction products analyzed by polyacrylamid-gel electrophoresis. Lane 1 shows the control L6S-G-L6S-G. Lane2 shows that L6S-G-L6S-G is degraded to L6S-G by PorA. Lane 3 shows that L6S-G-L6S-G is also degraded to L6S-G by PorB. The lower migration velocity of the L6S-G product is provoked by the sieve effect of the gel, because of the low size and high charge for this oligosaccharide the charge becomes more important for migration than the molecular weight (the effect of such a turning point is further demonstrated in the supplementary Figure 5 below).



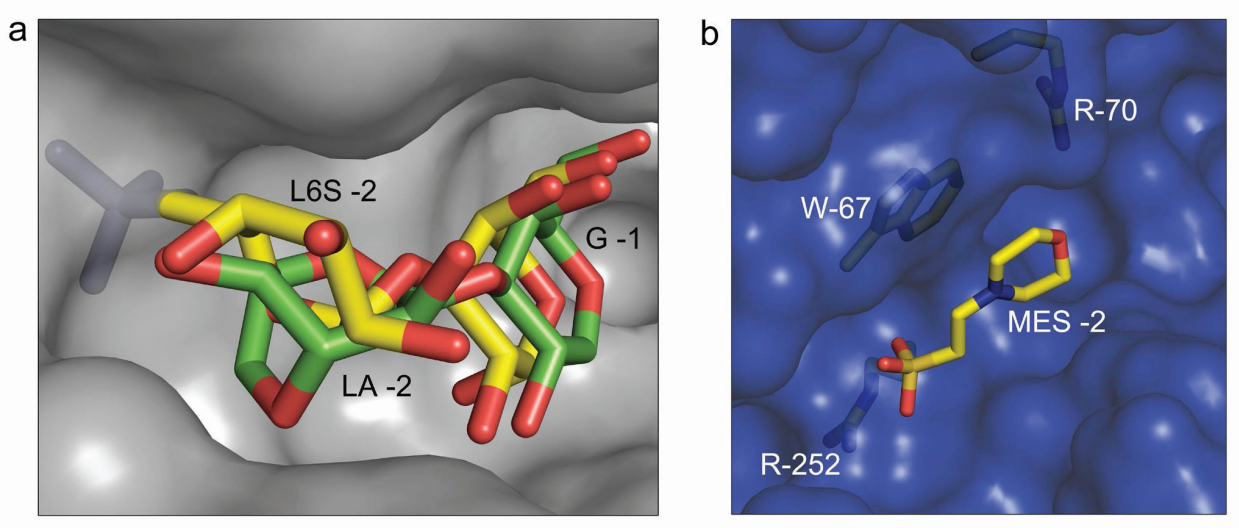
Supplementary Figure 5: **Phorphyran hydrolysis by PorA**. **(a)** Natural Porphyran was degraded with PorA. During the time course of degradation aliquots were analyzed by analytical size exclusion chromatography (SEC). PorA produces predominantly L6S-G corresponding to the peak at the end of the elution profile (~120 min). PorB similarly produces L6S-G as major final product (data not shown). After complete degradation of the polysaccharide the released oligosaccharides were purified by preparative SEC that resulted in a chromatogram reminiscent of the one shown

here. **(b)** The purified oligosaccharides (preparative SEC, chromatogram not shown) were analyzed by fluorophore assisted polyacrylamide-gelelectrophoresis with (2-amnoacridone, AMAC) as label. Lanes 1-4 correspond to here not further characterized oligosaccharides with degree of polymerization over 4 (>DP4). Lane 5 shows the DP4 of structure L6S-G-L6S-G as revealed by 1H-NMR (see below) and by cocrystallisation. Lanes 7-8 show the major product of both porphyranases L6S-G which corresponds to the final peak in the SEC chromatogram. Note that the lower velocity of L6S-G compared to the DP4 L6S-G-L6S-G oligosaccharide is due to a turning point in the influence of charge and mass on migration. **(c)** This effect is abolished when ANTS is used as label. In this case the L6S-G disaccharide shown in Lane 8 runs faster than the L6S-G-L6S-G tetrasaccharide on lane 6-7, as one would expect due to its smaller mass.



Supplementary Figure 6: **^{1}H-NMR analysis of oligosaccharides produced by porphyranases in comparison to agarose tetrasaccharide. (a) neo-agarotetraose (LA-G-LA-G); (b) neo-**

porphyrotetraose (L6S-G-L6S-G); (c) neo-porphyrodiose (L6S-G). For all spectra the α-anomeric configuration of D-galactose (reducing end) was the most downfield shifted signal at 5.3 ppm (J_{H1H2}= 3.8 Hz), whereas the signal for the β-anomeric configuration was at 4.64 ppm (J_{H1H2}= 7.7 Hz). The anomeric proton signals of L-galactose units allowed unambiguous assignment of the shifts from the 3,6-anhydro-α-L-galactose (LA) of the agarose motive at 5.17 ppm and the α-L-galactopyranose-6-sulfate (L6S) of the porphyran motive at 5.25 ppm. **(b)** The spectrum of the L6S-G-L6S-G oligosaccharide showed 2 additional signals at 4.49 and 4.41 ppm, compared to the disaccharide **(c)**, that were respectively ascribed to the internal H1 and H3 of the G residue. **(a)** Logically these signals of an internal D-galactose unit were present in the neo-agarotetraose spectrum as well. For both porphyran oligosaccharides the coupling constant of L6S-H1 was 3.9 Hz which is typical of the α-anomeric configuration. Moreover the HMBC experiments highlighted a correlation between L6S-C1 and Gr-H3 suggesting that the two residues were linked by α (1→3) linkage. All together, these results were consistent with a L6S-G disaccharide structure obtained by the cleavage of the β (1→4) glycosidic bond of porphyran.



Supplementary Figure 7: **Crystal structures of PorA and PorB show substrate specificity of porphyranases as compared to agarases. (a)** The subsite -2 pocket of PorA (Fig. 2a) is absent in agarase (AgaA) where the sulfate group of the modeled porphyran repeating unit leads to a sterical clash with the enzyme structure. The superposition illustrates how the 3,6-anhydro-bond locks the LA pyranose ring in the energetically unfavoured $^{1}C_{4}$ conformation that switches to $^{4}C_{1}$ by sulfation on C6 (L6S), explaining the different backbone structures of agarose and porphyran. **(b)** A bound MES molecule mimics the natural L6S at subsite -2 in PorB, indicating a PorA-like porphyran orientation and shows that the sulfonic group is stabilized by residue R-252.